

IoT Homework 2025/2026

Integrated Report: IEEE 802.15.4

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Interpretation used in this report: Exercise 1 is solved by sizing the Beacon Interval from the minimum output rate. The minimum case generates one 128 B packet every 2.5 s, so $BI = 2.5$ s. The 4 KB worst-case report is therefore spread over four BIs, requiring 8 CFP slots per node per BI. Because the network is beacon-enabled, the duty-cycle calculation explicitly adds one beacon slot to the CFP active part.

Executive Summary

Item	Main result
Exercise 1 output-rate PMF	$P(r_0)=0.6703$, $P(r_1)=0.2681$, $P(r_2)=0.0616$
Exercise 1 CFP allocation per BI	8 slots/node/BI, 24 CFP slots total per BI
Exercise 1 duty cycle	$0.04096 = 4.096\%$ per BI including one beacon slot
Exercise 1 scalability	Maximum total nodes = 7; additional nodes = 4
Exercise 2 energy-aware redesign	State-based beaconing using worn detection and motion detection
Exercise 2 BLE traffic saving	About 83.9% fewer BLE advertising events under the stated daily-use assumptions
Exercise 3 best initial frame size	$r_1 = 4$, $\eta_{\max} = 0.4533$

Exercise 1 - IEEE 802.15.4 Beacon-Enabled Network

The system contains one PAN coordinator and three ESP32 vibration-monitoring nodes. Each node processes a 10 s reporting window. The payload generated by a node depends on the number of detected vibration peaks X , with X following a Poisson distribution with $\lambda = 0.4$ peaks/window. The nominal bit rate is $R = 250$ kbps and each packet has length $L = 128$ bytes = 1024 bits. One packet fits exactly into one CFP slot.

Parameter	Value
Number of nodes M	3 vibration-monitoring nodes + 1 PAN coordinator
Reporting period T_p	10 s
Peak-count model	$X \sim \text{Poisson}(\lambda = 0.4)$
IEEE 802.15.4 bit rate R	250 kbps
Packet size L	128 bytes = 1024 bits
Slot assumption	1 packet = 1 CFP slot

1.1 Probability Mass Function of the output rate

The Poisson probabilities are:

- $P(X = 0) = e^{(-0.4)} = 0.6703$
- $P(X = 1) = 0.4 e^{(-0.4)} = 0.2681$
- $P(X > 1) = 1 - P(X = 0) - P(X = 1) = 0.0616$

The corresponding rates are computed by dividing the payload size in bits by the 10 s reporting period:

Event	Payload	Packets/window	Output rate	Probability
$X = 0$	512 B	4	409.6 bps	0.6703
$X = 1$	2 KB = 2048 B	16	1638.4 bps	0.2681
$X > 1$	4 KB = 4096 B	32	3276.8 bps	0.0616

1.2 Beacon Interval and consistent CFP slot assignment

First, the Beacon Interval is sized from the minimum output rate. In the minimum case, the node generates 512 B every 10 s. Since each packet is 128 B, this corresponds to 4 packets in 10 s, or one packet every 2.5 s.

$$BI = L / r_{\min} = (128 * 8) / 409.6 = 2.5 \text{ s.}$$

Therefore, one reporting period contains $T_p/BI = 10/2.5 = 4$ Beacon Intervals.

For a consistent CFP assignment, the CFP is dimensioned for the largest possible message, because the case $X > 1$ has nonzero probability. The largest message is 4 KB, which corresponds to $4096/128 = 32$ packets per node per reporting period.

Spread over 4 consecutive BIs, this requires $32/4 = 8$ slots per node per BI.

With $M = 3$ nodes, the total CFP allocation is $N_{\text{CFP}} = 3 * 8 = 24$ CFP slots per BI.

Quantity	Value
Beacon Interval BI	2.5 s
BIs per reporting period	4
Worst-case packets/node/period	32 packets
Nslots per node per BI	8 slots
N_{CFP} total per BI	24 slots

1.3 Slot time, active time, inactive time, and duty cycle

The duration of one CFP slot is $T_{\text{slot}} = L/R = 1024 / 250000 = 4.096$ ms.

The CFP-only time per BI is $T_{\text{CFP}} = 24 * 4.096$ ms = 98.304 ms.

Because the network is beacon-enabled, the active part contains one beacon slot plus the CFP: $N_{\text{active}} = 1 + 24 = 25$ slots.

$T_{\text{active}} = 25 * 4.096$ ms = 102.400 ms.

$T_{\text{inactive}} = \text{BI} - T_{\text{active}} = 2.5 \text{ s} - 0.1024 \text{ s} = 2.3976 \text{ s}$.

Duty cycle = $T_{\text{active}}/\text{BI} = 0.1024/2.5 = 0.04096 = 4.096\%$.

Quantity	Value
T_{slot}	4.096 ms
Nslots per node per BI	8 slots
N_{CFP} in the CFP	24 slots
T_{CFP}	98.304 ms
T_{active} including beacon	102.400 ms
T_{inactive}	2.3976 s
Duty cycle	$0.04096 = 4.096\%$

1.4 Additional nodes under a 10% duty-cycle limit

The maximum allowed active time per BI is $0.10 * 2.5 \text{ s} = 0.250 \text{ s}$. In slot units, this is $\text{floor}(0.250 / 0.004096) = 61$ active slots.

One active slot is reserved for the beacon, so the maximum number of CFP slots is 60.

Each node requires 8 CFP slots per BI. Therefore:

$M_{\text{max}} = \text{floor}(60 / 8) = 7$ total vibration-monitoring nodes.

Additional nodes = $7 - 3 = 4$.

Check: 7 nodes require $1 + 7*8 = 57$ active slots, giving $9.34\% < 10\%$. With 8 nodes, the active part becomes 65 slots, giving $10.65\% > 10\%$.

Final Numerical Values for Form Entry

Quantity	Value
Exercise 1 - P(r0)	0.6703
Exercise 1 - P(r1)	0.2681
Exercise 1 - P(r2)	0.0616
Exercise 1 - r0, r1, r2	409.6 bps, 1638.4 bps, 3276.8 bps
Exercise 1 - Beacon Interval	2.5 s
Exercise 1 - Nslots per node per BI	8
Exercise 1 - Nslots in CFP	24
Exercise 1 - Ts	4.096 ms
Exercise 1 - Tactive including beacon	102.400 ms
Exercise 1 - Tinactive	2.3976 s
Exercise 1 - duty cycle	0.04096 (4.096%)
Exercise 1 - additional nodes	4